

Next Analysis Plan

Variance Decomposition, Diagonal Projections, PCA, Low-Rank Variants, Heatmaps, and
Feature Decisions

Bayesian Football Analytics Pipeline

Planned from current fitted results

Abstract

This document is a planning note only. It does not propose immediate code changes. The goal is to define the next analytical work needed after the current real-data diagonal fit, simulation recovery checks, and validation outputs. The plan focuses on variance decomposition, projections using Σ_a and Σ_b , PCA-style summaries, additional low-rank simulation variants, sign-aware heatmaps of simulated effects, residual-sorted explained-variance summaries, and a decision framework for retaining or excluding `rate_*` variables.

1 Current Starting Point

The current validated pipeline has three main fitted objects:

- real-data diagonal additive model;
- simulated-data diagonal additive model fitted to low-rank simulated data;
- simulated-data low-rank recovery model.

The main existing conclusion is that the diagonal additive model is suitable for descriptive variance decomposition, while the current full low-rank recovery model has poor loading-level convergence. The next work should therefore separate two goals:

1. strengthen the thesis-facing interpretation of the diagonal model;
2. test narrower low-rank variants to understand whether the failure is caused by player covariance, season covariance, their combination, or limited season information.

2 Main Questions To Answer

Question	Why it matters	Best summary
Which features are mostly player, season, or residual driven?	This is the main thesis interpretation of the additive model.	Variance-share table and stacked bar plot sorted by residual share.
What do diagonal Σ_a and Σ_b imply geometrically?	Diagonal covariance matrices do not create rotated latent dimensions, but they still define feature-wise axes of player and season variation.	Feature ranking, diagonal projection scores, and player/season dominance maps.
Does PCA add information beyond diagonal variance ranking?	For diagonal matrices, PCA is trivial; for empirical effects or low-rank covariances, PCA can reveal correlated feature profiles.	Compare PCA on empirical/posterior effects and on low-rank covariance estimates.
Where exactly does low-rank recovery fail?	The current low-rank model mixes poorly; isolated experiments can diagnose whether the issue is player block, season block, or both.	Simulation recovery table across model variants.
How should signs of simulated player and season effects be shown?	Heatmaps can be misleading unless factor signs are aligned and rows are ordered sensibly.	Sign-aligned heatmaps for loadings/effects plus clustered summaries.
Should <code>rate_*</code> variables remain in the model?	Many rate variables are residual dominated, potentially noisy, and denominator sensitive.	Rate-variable audit and optional sensitivity comparison with rates removed.

3 Work Package 1: Variance Decomposition

3.1 Objective

Create a thesis-ready variance-decomposition summary for every fitted model where variance components are available. For each feature p , compute:

$$\begin{aligned} \text{player share}_p &= \frac{\sigma_{a,p}^2}{\sigma_{a,p}^2 + \sigma_{b,p}^2 + \sigma_{e,p}^2}, \\ \text{season share}_p &= \frac{\sigma_{b,p}^2}{\sigma_{a,p}^2 + \sigma_{b,p}^2 + \sigma_{e,p}^2}, \\ \text{residual share}_p &= \frac{\sigma_{e,p}^2}{\sigma_{a,p}^2 + \sigma_{b,p}^2 + \sigma_{e,p}^2}. \end{aligned}$$

3.2 Recommended Summaries

1. **All-feature variance table:** one row per feature with posterior mean player, season, residual, total variance, and the three shares.
2. **Residual-sorted table:** same table sorted descending by residual share. This directly identifies features least explained by stable player or season effects.
3. **Player-sorted and season-sorted tables:** top 15 features by player share and top 15 by season share.
4. **Stacked bar plot:** one horizontal bar per feature, sorted by residual share, with player, season, and residual colors.
5. **Feature-family summary:** compare `per90_*` versus `rate_*` variables using median and interquartile range of player, season, and residual shares.

3.3 Interpretation Rules

Pattern	Interpretation
High player share	The feature is stable across observed seasons for players; it is suitable for player-profile interpretation.
High season share	The feature is sensitive to common season-level shifts, possibly tactical, league-wide, measurement, or era effects.
High residual share	The feature is noisy or situational at player-season level after accounting for player and season.
High residual share for rates	Check denominator instability, rare events, and whether the rate should be retained.

4 Work Package 2: Projections Using Diagonal Σ_a And Σ_b

4.1 Key Point

For the diagonal model, Σ_a and Σ_b are diagonal matrices. Therefore they do not define rotated multivariate directions. Their eigenvectors are the original features, and their eigenvalues are the feature-specific variances. The right interpretation is not a latent PCA rotation, but a feature-axis weighting.

4.2 Planned Projections

1. **Player-variance projection:** weight each feature by $\sigma_{a,p}^2$ or by player share. This highlights the part of each observation most aligned with stable player variation.
2. **Season-variance projection:** weight each feature by $\sigma_{b,p}^2$ or by season share. This highlights features most aligned with season-level shifts.

3. **Residual projection:** weight each feature by residual share. This highlights noisy or weakly explained features.
4. **Player-versus-season contrast:** compute a feature contrast such as $\log(\sigma_{a,p}^2/\sigma_{b,p}^2)$, with a small stabilization constant if needed. This ranks features by whether they are more player-like or season-like.

4.3 Recommended Outputs

Output	Purpose
Diagonal variance scree plots for Σ_a and Σ_b	Show whether a few features dominate player or season variance.
Feature-axis projection table	Rank features by player, season, residual, and player-season contrast.
Two-dimensional map: player share vs season share	Identify player-dominant, season-dominant, residual-dominant, and mixed features.
Feature-family overlay	Show whether <code>rate_*</code> variables cluster in the residual-dominant area.

5 Work Package 3: PCA

5.1 What PCA Should And Should Not Mean Here

PCA on a diagonal covariance matrix is mostly a sorted variance ranking. It does not reveal cross-feature structure because off-diagonal covariance is zero. Therefore PCA should be used in three more informative places:

1. **PCA on posterior player effects** a_i from the diagonal model. This asks whether estimated player profiles vary along interpretable multifeature dimensions.
2. **PCA on posterior season effects** b_j from the diagonal model. This asks whether seasons separate along common tactical or measurement patterns.
3. **PCA/eigendecomposition of low-rank covariance estimates** in models where $\Sigma_a = \Lambda_a \Lambda_a^\top + \Psi_a$ or $\Sigma_b = \Lambda_b \Lambda_b^\top + \Psi_b$. This asks whether the low-rank structure creates interpretable shared dimensions.

5.2 Recommended PCA Summaries

Summary	Why it makes sense
Explained variance ratio by PC	Shows whether player or season effects have low-dimensional structure.

Summary	Why it makes sense
Top positive and negative loadings per PC	Makes each PC interpretable in football terms.
PC score scatterplots for players	Shows whether player profiles occupy meaningful regions; optionally label only extreme players to avoid clutter.
PC score plot for seasons	Only 12 seasons exist, so all seasons can be labelled.
Bootstrap/posterior uncertainty for loadings	Prevents overinterpreting unstable PCA directions.

6 Work Package 4: New Low-Rank Simulation Variants

6.1 Objective

The current low-rank recovery model is too broad to diagnose the failure precisely. The next experiments should isolate player and season covariance complexity.

6.2 Recommended Experiment Order

Experiment	Model structure	Reason
LR-A0	a_i low-rank, no season effect b_j	Start with the simplest player-only low-rank recovery. This checks whether player low-rank structure is identifiable without season competition.
LR-A1	a_i low-rank, b_j diagonal	Adds season adjustment while keeping season covariance simple. This tests whether player low-rank recovery survives realistic season controls.
LR-B1	a_i diagonal, b_j low-rank	Tests whether season low-rank structure is recoverable at all with only 12 seasons. This is expected to be difficult.
LR-AB	a_i low-rank, b_j low-rank	Current richest structure. Revisit only after the simpler cases are understood.

6.3 Recovery Metrics For Every Variant

Each simulation variant should produce the same recovery table:

- sampler health: divergences, treedepth hits, E-BFMI, acceptance;
- posterior health: max Rhat, number of Rhat warnings, min ESS;

- loading recovery: bias, MAE, RMSE, correlation, 90 percent coverage;
- covariance diagonal recovery: same metrics for Σ_a and/or Σ_b ;
- residual scale recovery;
- feature-level recovery sorted by worst absolute error.

6.4 Decision Logic

Outcome	Interpretation
LR-A0 works but LR-A1 fails	Season adjustment interferes with player low-rank identification.
LR-A1 works but LR-B1 fails	Player low-rank is feasible; season low-rank is not reliable with 12 seasons.
LR-A0 fails	The player low-rank specification or constraints need stronger identifiability before adding seasons.
LR-B1 fails but diagonal season works	Keep season diagonal in thesis models.
LR-AB works only with strong anchors or truth initialization	Treat low-rank as a sensitivity/recovery experiment, not as the main empirical model.

7 Work Package 5: Sign-Aware Heatmaps For Simulated a_i And b_j

7.1 Issue

Low-rank factor signs are not intrinsically identified. A factor can change sign if its loading and score both change sign. Therefore heatmaps of simulated or recovered factors need sign alignment before interpretation.

7.2 Recommended Heatmaps

1. **Truth heatmaps:** simulated true a_i and b_j effects by feature. These show the actual generated signal.
2. **Posterior mean heatmaps:** recovered posterior mean effects, aligned to truth where simulation truth exists.
3. **Error heatmaps:** posterior mean minus truth, using a diverging color scale centered at zero.
4. **Sign agreement heatmaps:** sign of posterior mean compared with sign of truth, useful when magnitude recovery and sign recovery differ.
5. **Uncertainty heatmaps:** posterior probability that an effect is positive, $P(a_{ip} > 0)$ or $P(b_{jp} > 0)$, where available.

7.3 Display Strategy

Showing every player and every feature in one figure may be too dense because there are 1,650 players and 53 features. The plan should include:

- full heatmap as an appendix artifact;
- clustered heatmap by player-effect profile;
- top 50 players by absolute total effect norm;
- repeated-player-only heatmap, because repeated observations identify player effects better;
- all 12 seasons labelled individually, because the season heatmap is small enough to read.

8 Work Package 6: Explained Variance Sorted By Residual

8.1 Objective

For every completed model where variance components are meaningful, produce a single comparable explained-variance table sorted by residual share.

8.2 Recommended Master Table

Feature	Family	Player	Season	Residual	Total	Flag
rate_successful_crosses	rate	–	–	high	–	review
rate_successful_sliding _tackles	rate	–	–	high	–	review
rate_shots_on_target	rate	–	–	high	–	review
rate_goal_conversion	rate	–	–	high	–	review

The actual table should contain all 53 features. The rows above illustrate the expected high-residual rate-variable cases from the current fitted results.

8.3 Useful Flags

- **player-dominant**: player share above a chosen threshold;
- **season-dominant**: season share above a chosen threshold;
- **residual-dominant**: residual share above a chosen threshold;
- **rate-review**: variable name starts with **rate_*** and has high residual share;
- **low-ESS**: variance component has weak posterior effective sample size;
- **Rhat-warning**: variance component has convergence warning.

9 Work Package 7: Decide Whether `rate_*` Variables Are Needed

9.1 Reason To Reconsider Rates

Rate variables can be statistically fragile because they depend on denominators. A player with few attempts can have an extreme rate that is not a stable trait. The current real-data model already suggests that several rate features are mostly residual variation.

9.2 Decision Framework

Criterion	Action
High residual share and rare denominator	Strong candidate for removal or separate modelling.
High player share and adequate denominator	Candidate to keep as a stable skill-like feature.
High correlation with a cleaner <code>per90_*</code> variable	Consider removing the noisier rate variable.
Important football interpretation despite noise	Keep, but report caveat.
Unstable under sensitivity analysis	Remove from main model or move to appendix.

9.3 Recommended Sensitivity Comparison

Do not remove all rates automatically. Instead, plan one sensitivity fit or analysis with:

- full feature set, current baseline;
- no `rate_*` variables;
- optional filtered-rate set, keeping only rates with adequate denominators and non-extreme residual share.

Compare these on:

- stability of top player-dominant `per90_*` features;
- posterior health;
- residual-share distribution;
- whether season-level conclusions change.

10 Prioritized Execution Plan

Priority	Task	Output
1	Create final variance-decomposition tables sorted by residual, player, and season share.	Thesis-ready CSV/table and stacked bar plot.
2	Add feature-family summaries for <code>per90_*</code> vs <code>rate_*</code> .	Rate-variable audit table.
3	Produce diagonal Σ_a , Σ_b , and residual projection summaries.	Feature-axis ranking and player-season contrast plot.
4	Run PCA summaries on posterior player and season effects, not merely on diagonal covariance matrices.	PC explained variance, loading tables, score plots.
5	Design and run low-rank simulation variants in order: LR-A0, LR-A1, LR-B1, then LR-AB only if justified.	Recovery comparison table.
6	Produce sign-aligned heatmaps for simulated truth, posterior means, errors, and sign agreement.	Heatmap appendix and selected readable panels.
7	Decide whether <code>rate_*</code> variables stay in the main thesis model.	Feature inclusion recommendation.

11 Recommended Final Summary Figures

1. **Variance decomposition by feature**, sorted by residual share. This should be the main thesis figure.
2. **Player share vs season share scatter**, with point shape or color for `per90_*` versus `rate_*`.
3. **Residual-share distribution by feature family**, comparing `per90_*` and `rate_*`.
4. **PCA loading table and score plot for player effects**, if PCs are stable.
5. **Season-effect heatmap**, all 12 seasons by selected features.
6. **Low-rank recovery dashboard**, one table comparing LR-A0, LR-A1, LR-B1, and LR-AB on convergence and recovery metrics.
7. **Simulation heatmap appendix**, truth, posterior mean, and error maps after sign alignment.

12 Expected Interpretation If Current Patterns Persist

If the current results remain stable, the likely final story is:

- the diagonal additive model is the main empirical model because it gives interpretable and reasonably stable variance decomposition;
- many `per90_*` volume and intensity variables are stable player-profile measures;
- several `rate_*` variables are mostly residual and should be treated cautiously;
- season-level effects exist for selected tactical or measurement-sensitive features, but broad season covariance is hard to estimate with only 12 seasons;
- low-rank player covariance may be feasible in simplified simulations, but low-rank season covariance is expected to be fragile;
- full low-rank player-plus-season recovery should not be the main thesis claim unless the simpler variants show reliable convergence and recovery.

13 Immediate Non-Coding Next Step

Before implementing any new model variant, define the final reporting tables and figures exactly. The first deliverable should be a compact specification listing each output, its columns, sorting rule, and interpretation sentence. That avoids running additional models before the thesis summaries are clear.